

VALLEYRAT PELOADER ARTIFACT HUNT

Exact file-metadata hunt for the AES-encrypted backdoor artifact staged by the source-observed QNWallpaper delivery chain. A match is not proof of memory loading, persistence, C2, or compromise.

HUNTING QUERY

CONFIDENCE: MEDIUM

REPORT DATE

1 September 2026

STATUS

Untested implementation sketch

PRIMARY TELEMETRY

Defender XDR /
DeviceFileEvents

ATT&CK

T1027.013

Campaign-specific artifact / IOC-based. Exact filename, path and MD5 provide specificity, not durability.

CONTROLLED EIGHT-PAGE THREAT DOSSIER

QNWallpaper 5.4.0.1662

random subdirectory



PeLoader

AES-encrypted backdoor file

EXACT MD5 MATCH

hunt, collect, corroborate



EXECUTIVE ASSESSMENT

Decision-grade summary and strict evidence boundary.

ASSESSMENT

OPERATIONAL VALUE

9.2

High-specificity exposure hunt for an exact encrypted backdoor file observed in a large software-bundling campaign.

CLASSIFICATION

Campaign-specific artifact / IOC-based

Hunting query - Untested implementation sketch. The logic is not durable enough for production alerting.

>100k source detections

>1.5k affected users

3 exact hunt conditions

0 production rules

Source fact

PeLoader is an AES-encrypted backdoor file deployed under the documented QNWallpaper version path.

Analytical meaning

Exact MD5 plus path and filename is suitable for exposure hunting and forensic collection.

Assumptions

Defender for Endpoint observed the file event and calculated MD5 before deletion or quarantine.

Promotion boundary

A match supports artifact presence. It does not prove the payload was decrypted, memory-loaded, executed, or connected to C2.



THREAT MECHANICS

Source-supported delivery chain and the observable boundary.

BEHAVIOR

01 Bundled installer

Adware-like installer deploys signed QNWallpaper and distracts the user.

→

02 Product directory

Files are unpacked under version 5.4.0.1662 and a random subdirectory.

→

03 Malicious libcef

QNWallpaper processes load a malicious co-located libcef.dll.

04

Encrypted file

PeLoader stores an AES-encrypted DLL payload.

→

05 Memory load

The loader decrypts and loads the DLL in memory.

→

06 ValleyRAT

Source analysis identifies the backdoor payload.

Behavior	Source support	Query coverage
PeLoader filename/path/MD5	Direct artifact table	Exact
Malicious libcef DLL loading	Reverse-engineered	Not correlated
AES decryption and memory loading	Reverse-engineered	Unavailable
Startup persistence	Reverse-engineered	Unavailable

ATT&CK

T1027.013 Encrypted/Encoded File applies to the source behavior because the backdoor DLL is stored AES-encrypted. File metadata alone does not reveal encryption.



TELEMETRY & EVIDENCE MODEL

Minimum data contract and observability gaps.

TELEMETRY

REQUIRED

DeviceFileEvents

Microsoft Defender for Endpoint file creation, modification, rename or other file-system events with calculable MD5.

OPTIONAL ENRICHMENT

DeviceImageLoadEvents, DeviceProcessEvents, DeviceRegistryEvents, file collection, signature/reputation data and network telemetry.

Field	Reliability	Purpose
Timestamp, DeviceId, DeviceName	Documented	Event scope
FileName, FolderPath, MD5	Documented	Exact hunt conditions
ActionType	Documented; values in portal	Activity context only
InitiatingProcess*	Documented	Delivery pivot
FileOrigin*	Documented; may be empty	Origin enrichment
ReportId	Documented	Use with DeviceName and Timestamp

Unavailable through this query

File content, AES state, in-memory DLL, DIIMain invocation, sideload success, startup execution, C2 traffic and user compromise.

Coverage failure

The file predates onboarding, no event was generated, it was inaccessible for hashing, the artifact was changed, or endpoint retention expired.



HUNTING LOGIC

Exact campaign artifact retrieval with no threshold or correlation.

KQL

HYPOTHESIS

If an attacker deploys the observed ValleyRAT package, file telemetry should show the exact PeLoader MD5 under the documented QNWallpaper version path. Legitimate overlap is reduced through the three exact conditions and resolved through origin, initiating process, file collection and adjacent artifacts.

KQL - UNTESTED HUNT

```
DeviceFileEvents
| where FileName == "PeLoader"
| where FolderPath startswith @"C:\Program Files\QNWallpaper\5.4.0.1662\"
| where MD5 =~ "48826d5ca845979d2e6ebd66dc1aae90"
| project Timestamp, DeviceId, DeviceName, ActionType,
    FileName, FolderPath, FileSize, MD5, SHA1, SHA256,
    FileOriginUrl, FileOriginReferrerUrl, FileOriginIP,
    InitiatingProcessFileName, InitiatingProcessFolderPath,
    InitiatingProcessCommandLine, InitiatingProcessMD5,
    InitiatingProcessSHA1, InitiatingProcessAccountName, ReportId
| order by Timestamp desc
```

THRESHOLD	WINDOW	JOIN	KEY	STATUS
None	Analyst-selected	None	Exact artifact	Untested

Why Defender XDR

The documented schema exposes the file name, folder and MD5 directly, plus initiating-process and origin pivots, without assuming a customer parser.



TRIAGE & INVESTIGATION

Turn an exact match into defensible endpoint evidence.

WORKFLOW

1

Validate the event

Confirm all three conditions and preserve Timestamp, DeviceName and ReportId.

2

Collect the file

Acquire the artifact where policy permits and re-calculate hashes.

3

Inspect the creator

Review initiating process, command line, account, origin and installer lineage.

4

Pivot locally

Look for the documented malicious libcef.dll, QNWallpaper processes and persistence.

5

Scope and contain

Search fleet-wide, validate memory/network evidence and isolate confirmed cases.

Escalate

Exact hash on a user endpoint, matching installer lineage, malicious libcef or memory/network corroboration.

Benign context

Approved malware lab, forensic repository, quarantine store or controlled test system.

Inconclusive

Hash missing, partial path, deleted file, or artifact visible only in a scanner cache.

Response boundary

Do not conclude ValleyRAT execution from a cached or quarantined file. Confirm endpoint role, file state and process or memory evidence.



FALSE POSITIVES & VALIDATION

What must be proven before operational use.

QUALITY CONTROL

SCENARIO	OVERLAP	DISPOSITION
Malware-analysis repository	M	Verify approved lab role and sample provenance
AV/EDR quarantine or cache	M	Confirm folder semantics and remediation state
MD5 collision	L	Collect file and confirm SHA1/SHA256 where available
Production user endpoint	H	Escalate for collection and adjacent evidence

Validation steps

1. Verify DeviceFileEvents retention and MD5 population.
2. Run against historical data and disposition every exact match.
3. Use an authorized inert copy to confirm event/hash coverage.
4. Validate origin and initiating-process fields on sample events.

Decision criteria

Success: exact test and production artifacts are retrievable with context.

Rejection: matches are dominated by approved repositories and cannot be scoped.

Failure: file events or MD5 are absent.

Tuning

Allowlist only verified lab/quarantine paths or device roles. Do not remove the exact hash/path conditions or alert on PeLoader alone.



CONCLUSION & REFERENCES

Coverage statement, cross-platform limits and document control.

FINAL

CONCLUSION

A precise exposure hunt, not a durable behavior detector.

The artifact and values are directly source-supported and Defender fields are documented. Environmental validation, hash availability and adjacent evidence remain required.

Cross-SIEM decision

Splunk: no reliable SPL without a verified EDR source, index, sourcetype and exact file/hash/path extractions.

Chronicle: no reliable YARA-L without a verified parser and UDM file-path/hash mapping.

Deferred ideas

Generic PeLoader alerting, path-only variants, legacy DisableAntiSpyware alerting, unverified startup-path logic and multi-table correlations.

References

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3. Microsoft Learn - DisableAntiSpyware, updated 29 Mar 2024.
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Document control

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